

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

1. Ans: Smart Home Automation System – Intelligent Agents

A smart home automation system manages lighting, temperature, and security by responding to real-time inputs such as motion detection, temperature changes, and time of day. It also learns user preferences to improve comfort and efficiency.

Types of Intelligent Agents

1. Simple Reflex Agent

- Works only on the current percept using condition-action rules.
- Does not remember previous states.

Example:

- If motion is detected → Turn ON lights.
- If no motion → Turn OFF lights.

Advantages

- Simple and fast.
- Easy to implement.

Disadvantages

- Cannot learn user preferences.
- Cannot handle complex situations.

2. Model-Based Reflex Agent

- Maintains an internal model of the environment.
- Uses current and previous information for decisions.

Example:

- Remembers that the room is occupied even if motion stops briefly.
- Adjusts lighting based on previous room status.

Advantages

- Better decision making.
- Handles partially observable environments.

Disadvantages

- More complex than simple reflex agents.
- Requires memory.

3. Goal-Based Agent

- Takes actions to achieve predefined goals.

Example Goals

- Maintain room temperature at 24°C.
- Keep home secure.
- Reduce unnecessary energy usage.

Advantages

- Makes intelligent decisions.
- Can compare multiple actions before choosing one.

Disadvantages

- Requires search and planning.
- Higher computational cost.

4. Utility-Based Agent

- Selects the action that provides the highest utility (best overall outcome).
- Balances multiple objectives simultaneously.

Example

The agent considers:

- Comfort
- Energy efficiency
- Security
- Electricity cost

It may reduce AC usage when electricity prices are high while maintaining acceptable comfort.

Advantages

- Produces the best overall performance.
- Handles conflicting goals effectively.

Disadvantages

- Utility function design is complex.
- Requires more computation.

Most Effective Approach

A hybrid approach combining Model-Based, Goal-Based, and Utility-Based agents is the most effective.

- **Model-Based Agent** remembers the home's current state.
- **Goal-Based Agent** ensures comfort and security.
- **Utility-Based Agent** optimizes energy consumption and user satisfaction.

This hybrid system provides:

- Improved comfort
- Better energy efficiency
- Enhanced security

- Adaptation to user preferences over time

2. Ans: Robot in an Unknown Maze – UCS and IDS

A robot is placed in an unknown maze and must find the exit without prior knowledge of the environment.

Uniform Cost Search (UCS)

Uniform Cost Search expands the node with the **lowest cumulative path cost**.

Working

1. Start from the initial position.
2. Store nodes in a priority queue.
3. Expand the node with the lowest cost.
4. Continue until the exit is reached.

Advantages

- Complete.
- Finds the optimal path when costs differ.
- Suitable for weighted paths.

Disadvantages

- High memory usage.
- Slower for large search spaces.

Complexity

- **Time:** $O(b^{(1 + C^*/\epsilon)})$
- **Space:** $O(b^{(1 + C^*/\epsilon)})$
where:
 - **b** = branching factor
 - **C*** = optimal solution cost
 - **ε** = minimum step cost

Iterative Deepening Search (IDS)

IDS repeatedly performs Depth-First Search with increasing depth limits.

Working

1. Search with depth limit 0.
2. Increase limit to 1.
3. Continue increasing until the goal is found.

Advantages

- Complete.
- Uses very little memory.

- Suitable when solution depth is unknown.

Disadvantages

- Repeats node expansions.
- Slower than BFS for shallow solutions.

Complexity

- **Time:** $O(b^d)$
- **Space:** $O(bd)$
where:
 - **b** = branching factor
 - **d** = depth of the solution

Relation to Intelligent Agents

Simple Reflex Agent

- Acts only on current observations.
- Not suitable for unknown mazes.

Model-Based Agent

- Maintains a map of explored paths.
- Avoids revisiting known locations.
- Suitable for maze navigation.

Goal-Based Agent

- Plans actions to reach the exit.
- Uses search algorithms such as UCS or IDS.

Utility-Based Agent

- Chooses paths that maximize utility.
- Considers distance, time, and energy consumption.

Best Combination

The most effective solution is:

- **Agent:** Model-Based Goal-Based Agent
- **Search Strategy:** Uniform Cost Search (UCS)

Justification

- The model-based component stores explored paths.
- The goal-based component plans toward the exit.
- UCS guarantees the lowest-cost path when movement costs vary.
- This combination provides efficient, reliable, and optimal navigation.

